PROJECT SUBMISSION

Nand to Tetris

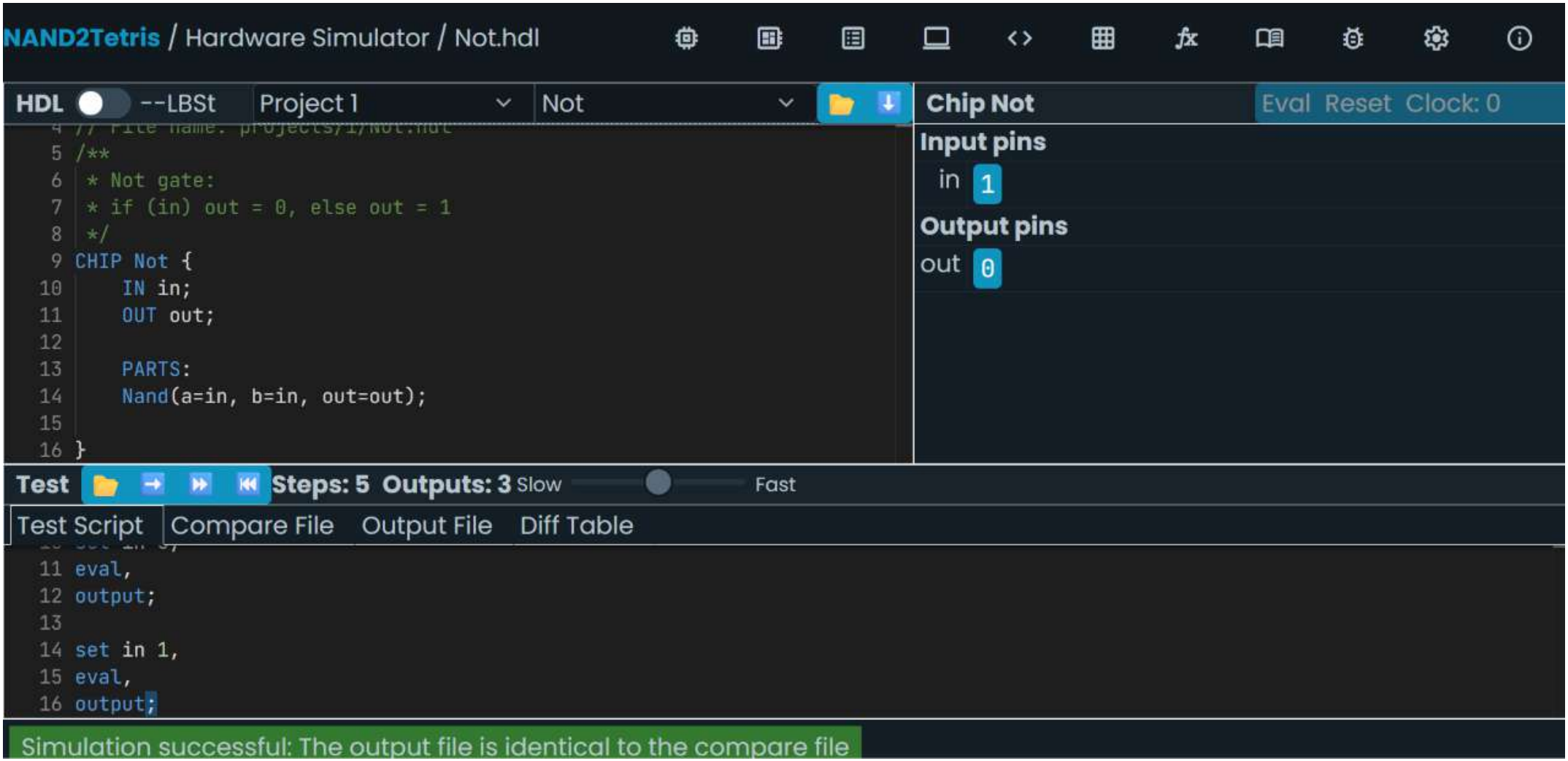
Hardware Simulator Screenshots

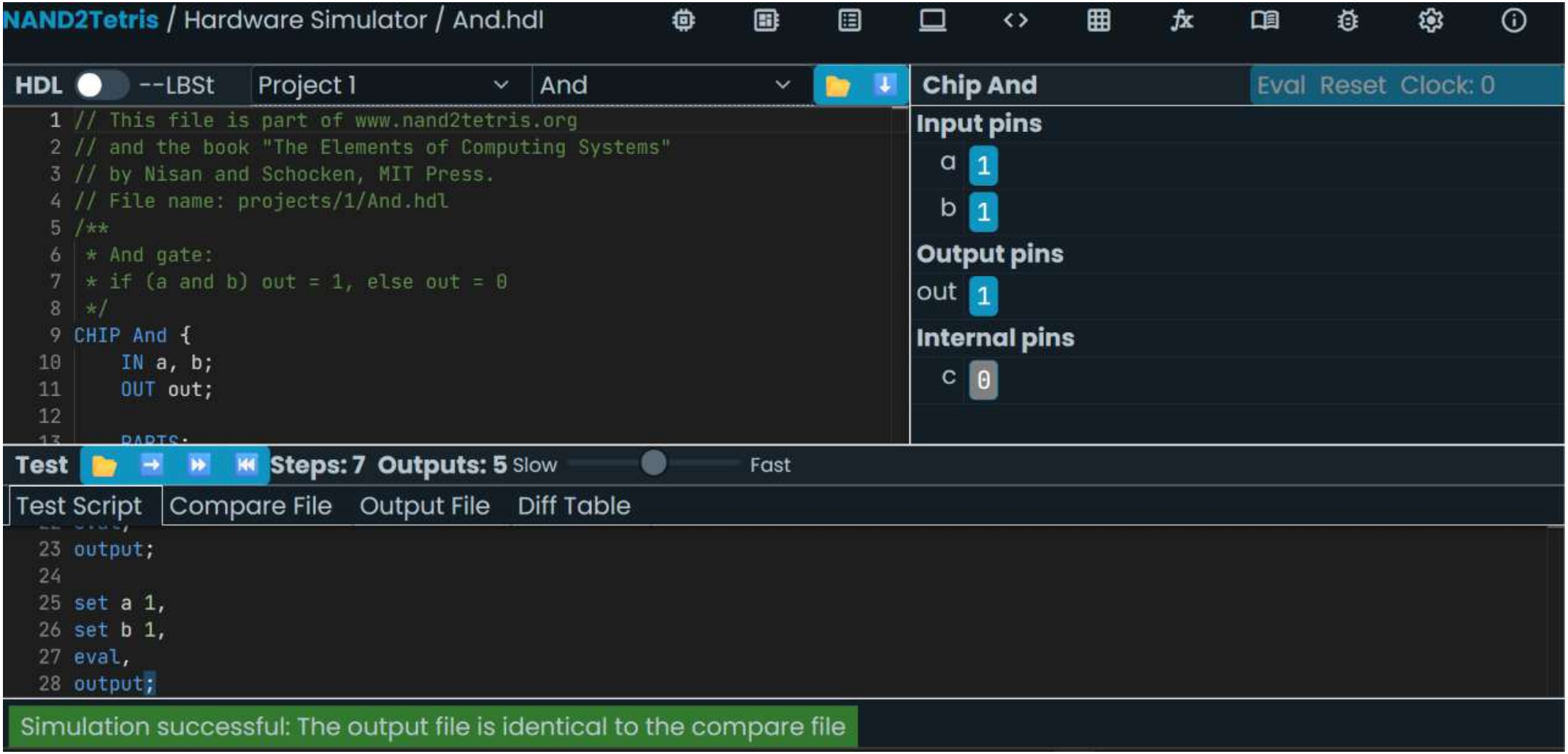
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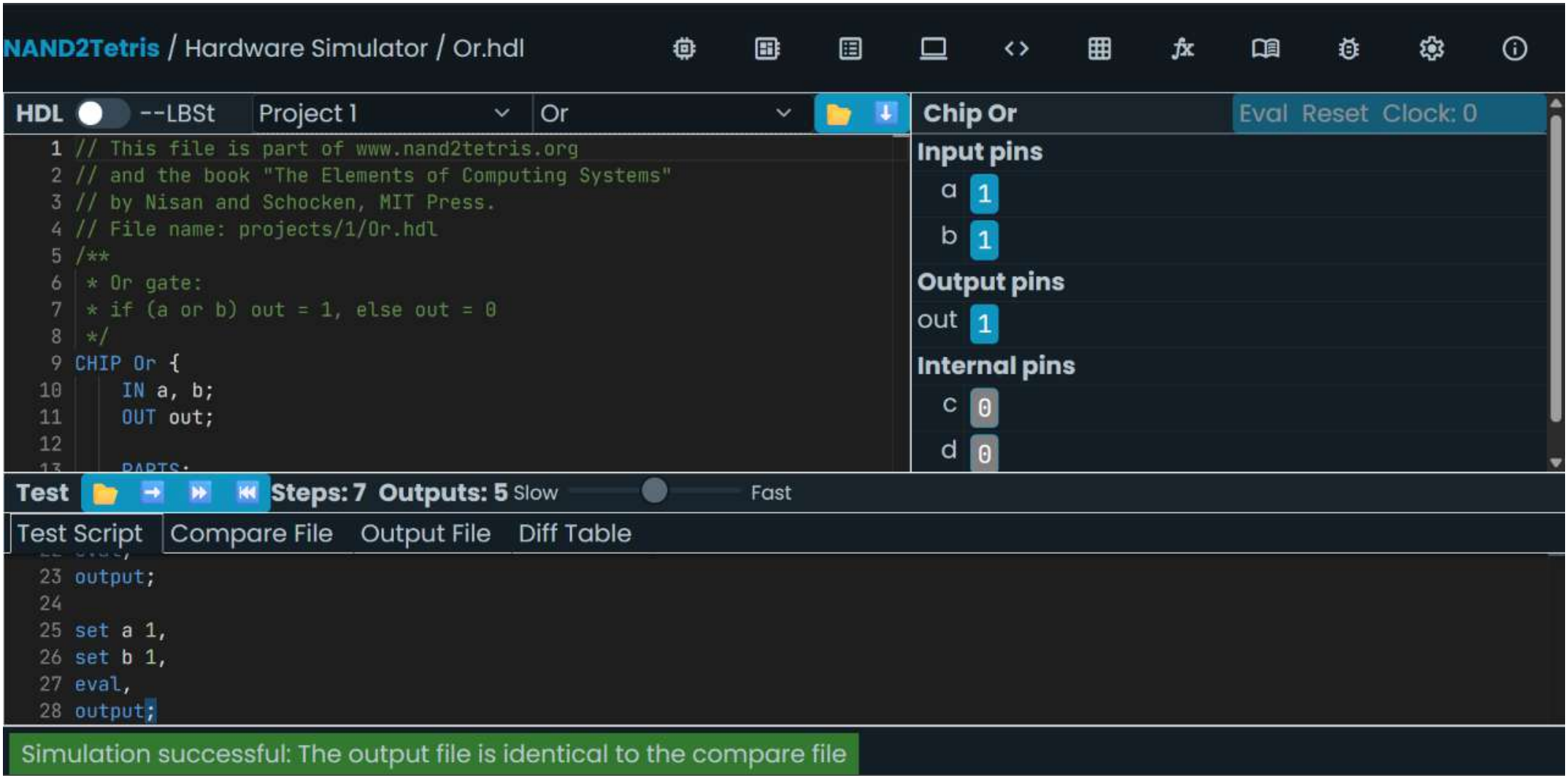
Satakshi Gupta

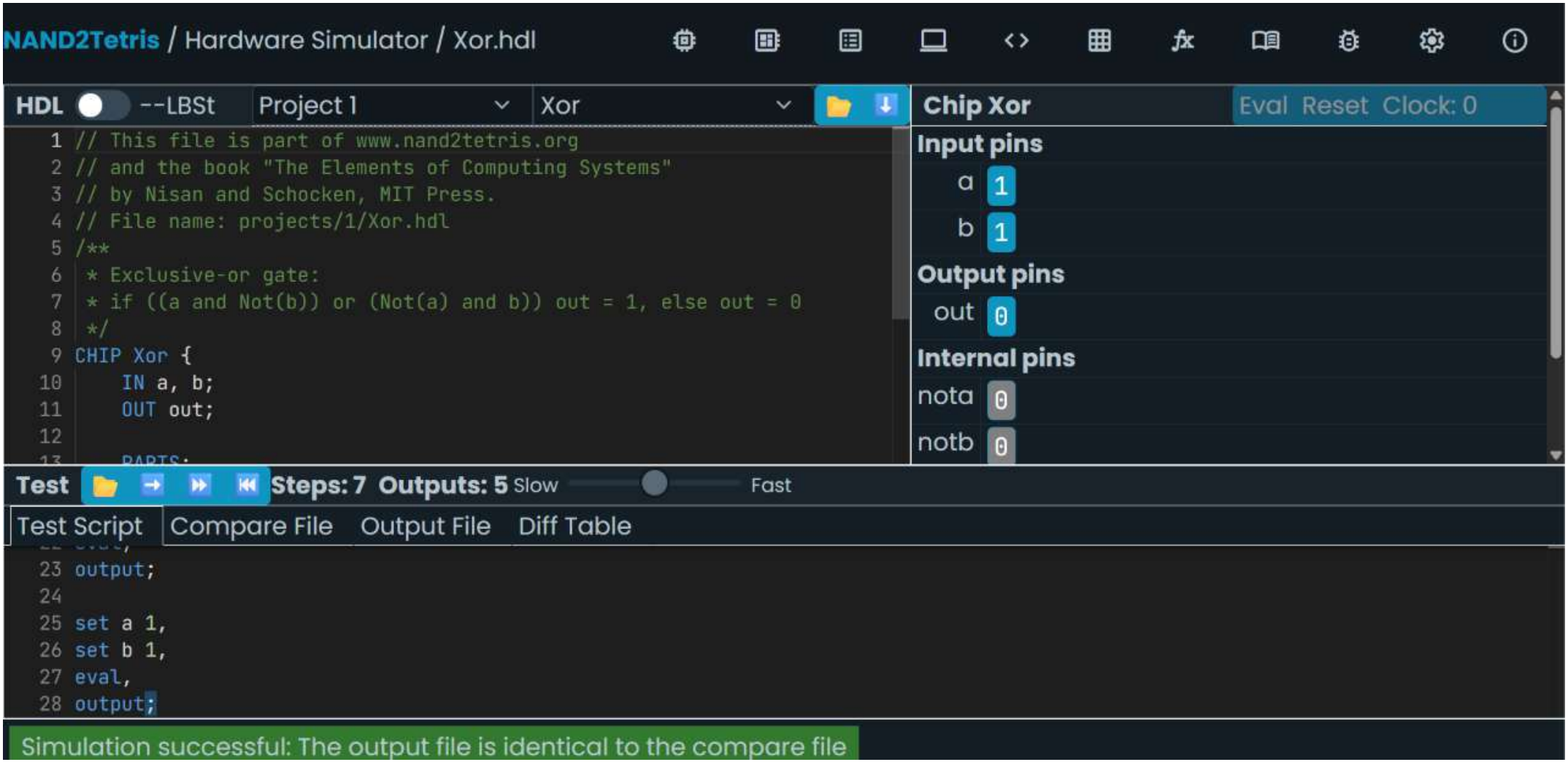
ROLL NO

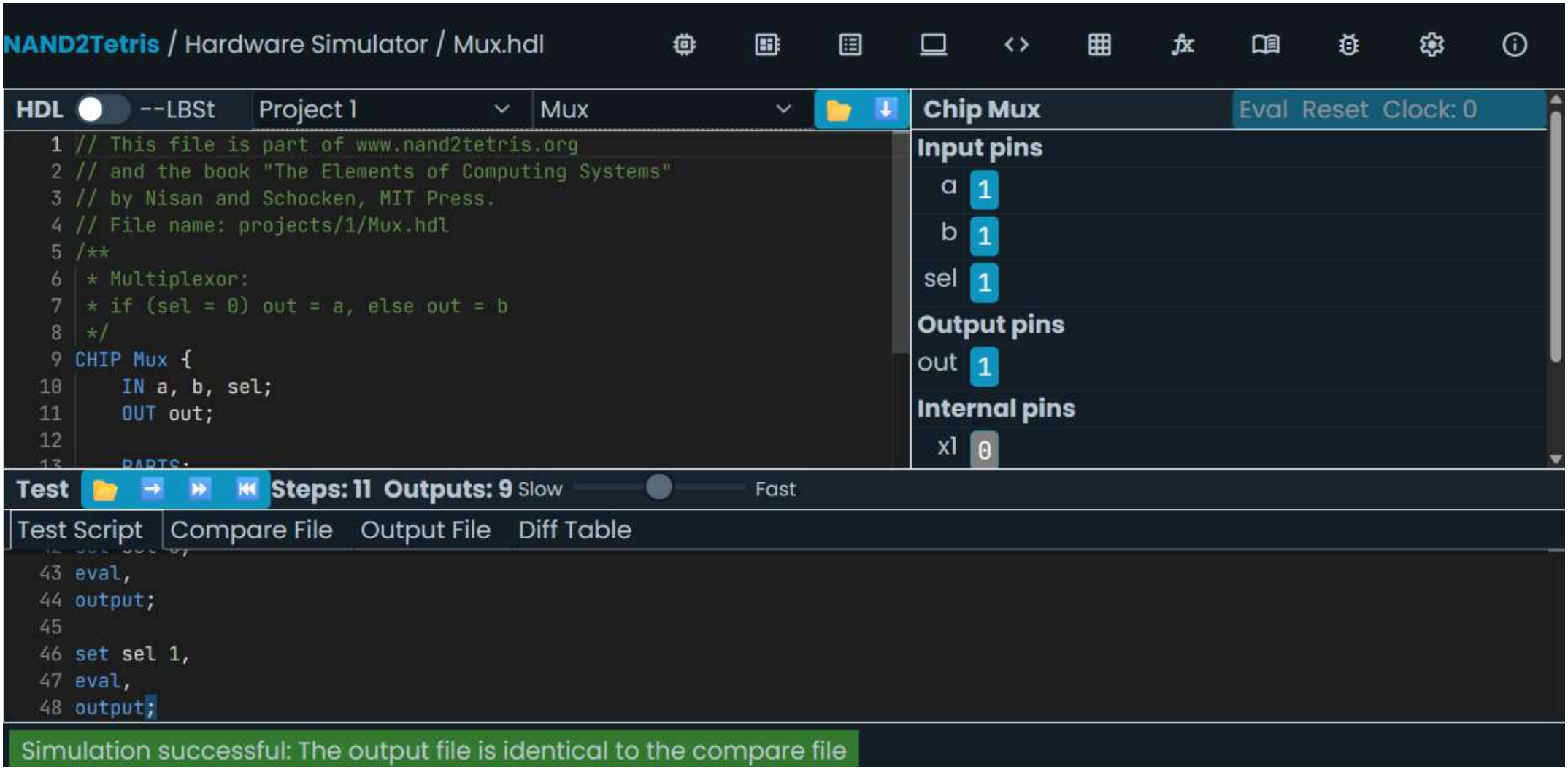
2501730461

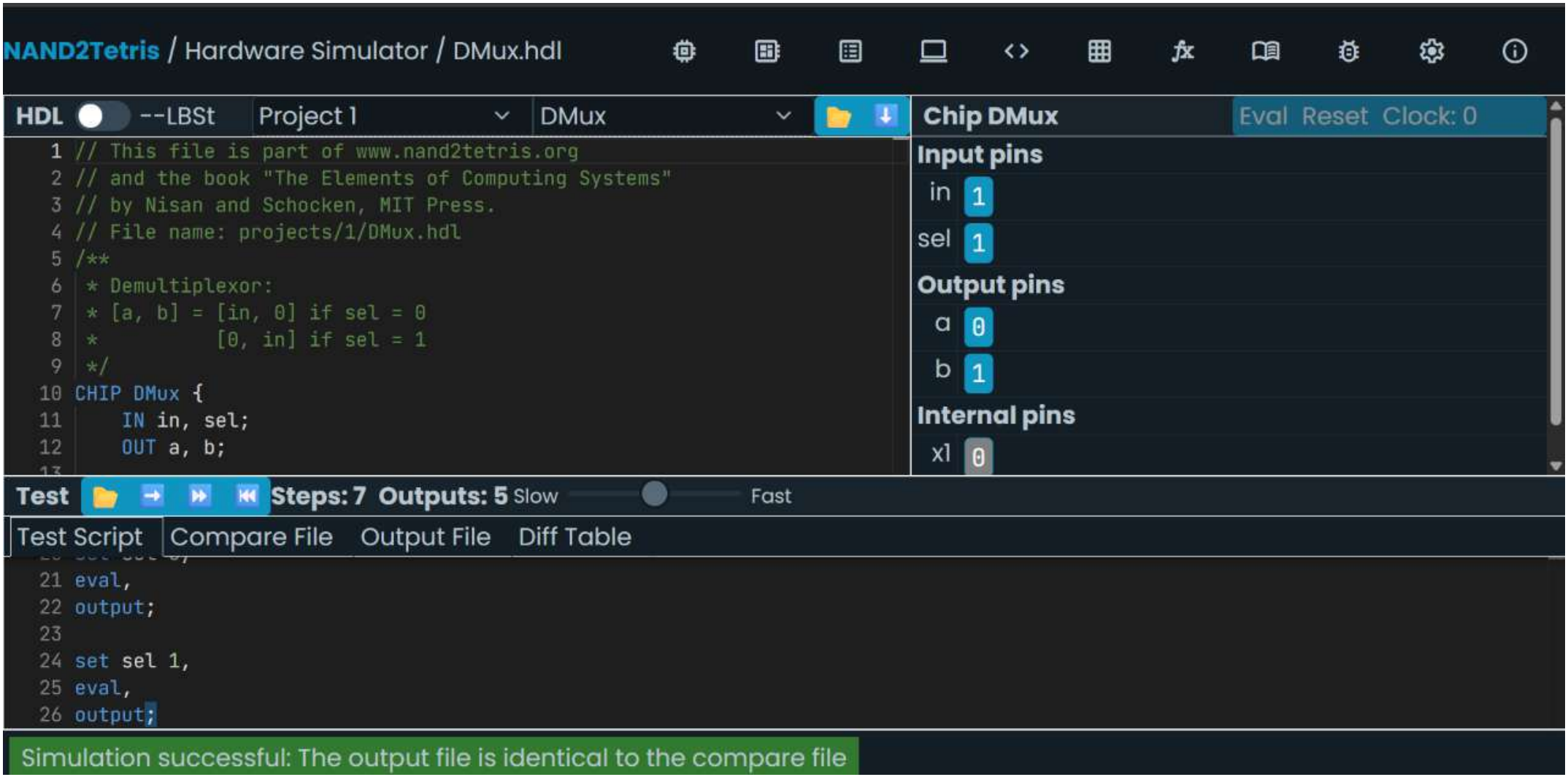

NAND2Tetris / Hardware Simulator / Not.hdl
HDL --LBSt
Project 1
Not
// File name: projects/1/Not.hdl
/**
* Not gate:
* if (in) out = 0, else out = 1
*/
CHIP Not {
IN in;
OUT out;
PARTS:
Nand(a=in, b=in, out=out);
}
Chip Not
Eval Reset Clock: 0
Input pins
in 1
Output pins
out 0
Test
Steps: 5 Outputs: 3
Slow
Fast
Test Script
Compare File
Output File
Diff Table
eval,
output;
set in 1,
eval,
output;
Simulation successful: The output file is identical to the compare file

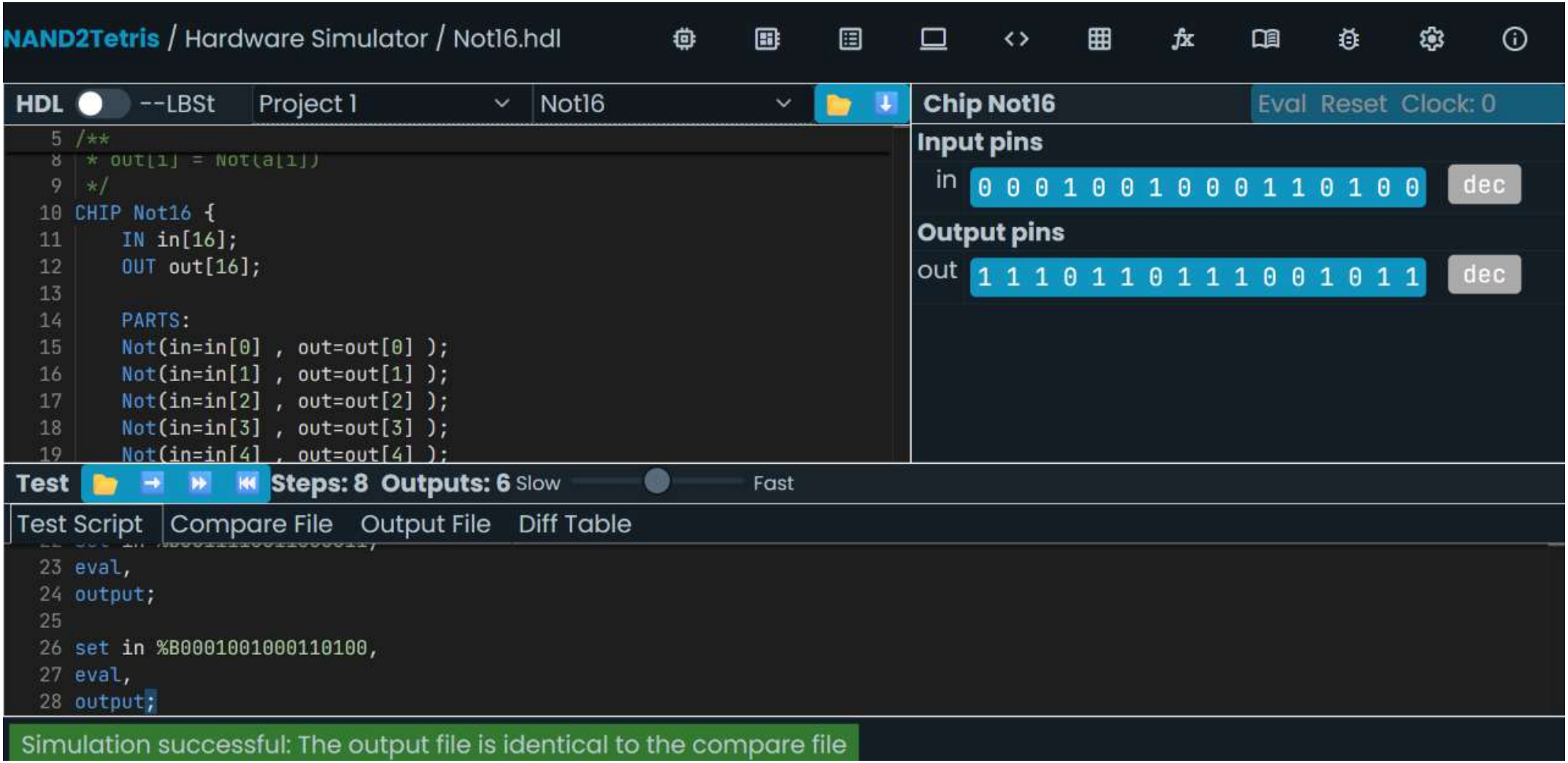

NAND2Tetris / Hardware Simulator / And.hdl
HDL --LBSt
Project 1
And
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/And.hdl
5 /**
6 * And gate:
7 * if (a and b) out = 1, else out = 0
8 */
9 CHIP And {
10 IN a, b;
11 OUT out;
Chip And
Eval Reset Clock: 0
Input pins
a 1
b 1
Output pins
out 1
Internal pins
c 0
Test
Steps: 7 Outputs: 5 Slow
Fast
Test Script
Compare File
Output File
Diff Table
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
28 output;
Simulation successful: The output file is identical to the compare file

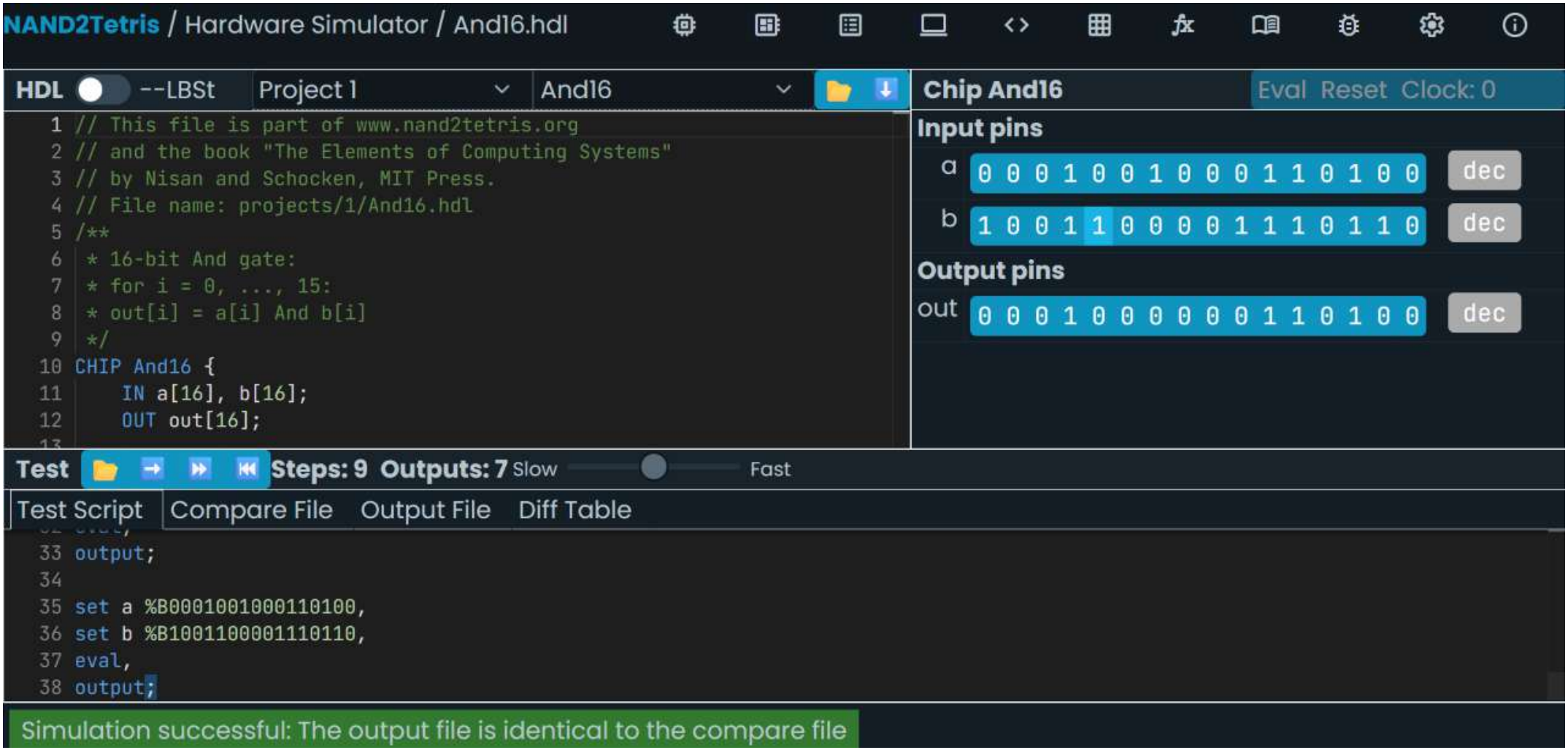

NAND2Tetris / Hardware Simulator / Or.hdl
HDL --LBSt Project 1 Or
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Or.hdl
5 /**
6 * Or gate:
7 * if (a or b) out = 1, else out = 0
8 */
9 CHIP Or {
10 IN a, b;
11 OUT out;
Chip Or
Eval Reset Clock: 0
Input pins
a 1
b 1
Output pins
out 1
Internal pins
c 0
d 0
Test
Steps: 7 Outputs: 5 Slow Fast
Test Script
Compare File
Output File
Diff Table
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
28 output;
Simulation successful: The output file is identical to the compare file

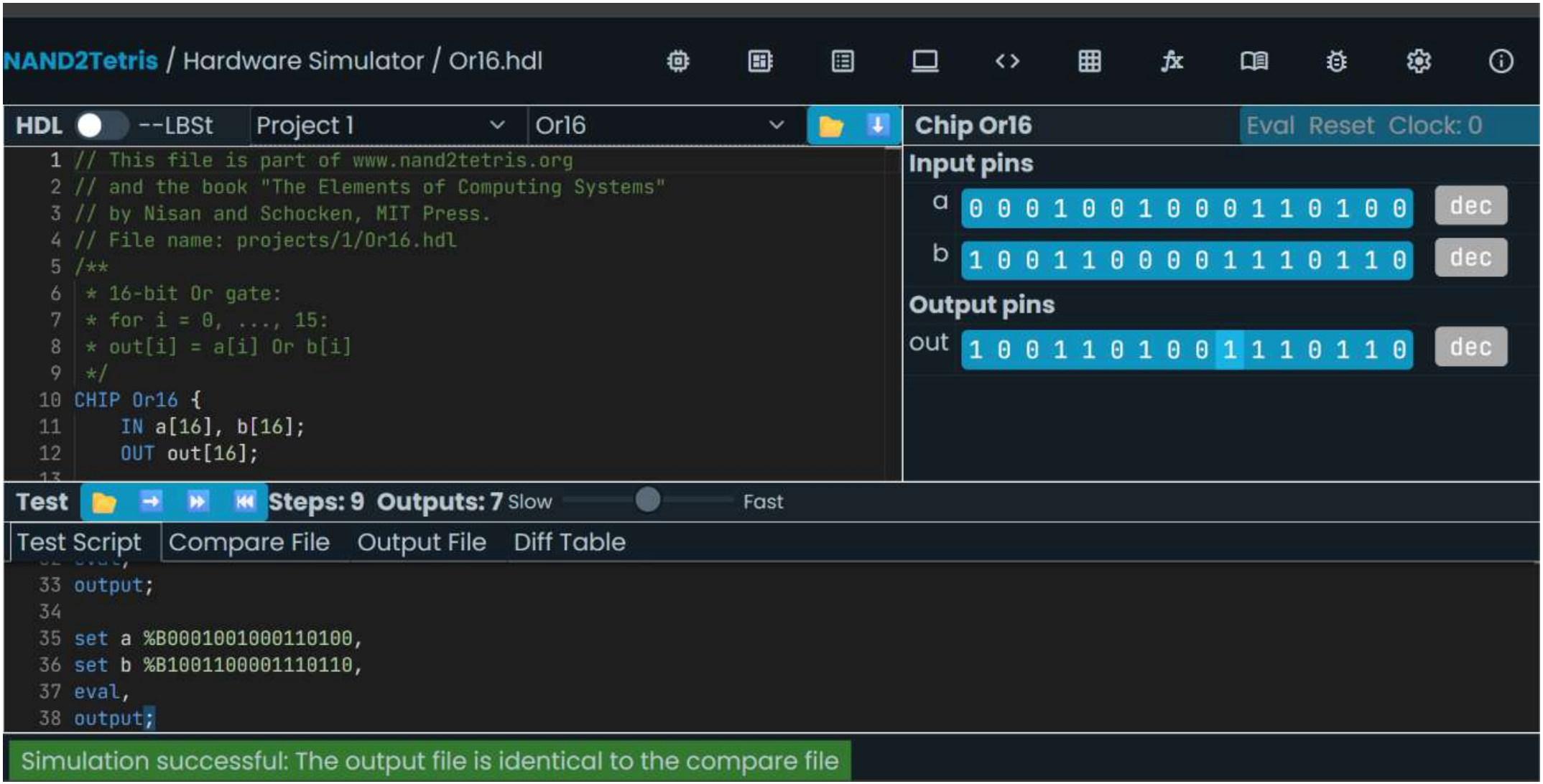

NAND2Tetris / Hardware Simulator / Xor.hdl
HDL --LBSt
Project 1
Xor
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Xor.hdl
5 /**
6 * Exclusive-or gate:
7 * if ((a and Not(b)) or (Not(a) and b)) out = 1, else out = 0
8 */
9 CHIP Xor {
10 IN a, b;
11 OUT out;
Chip Xor
Eval Reset Clock: 0
Input pins
a 1
b 1
Output pins
out 0
Internal pins
nota 0
notb 0
Test
Steps: 7 Outputs: 5 Slow
Fast
Test Script
Compare File
Output File
Diff Table
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
28 output;
Simulation successful: The output file is identical to the compare file

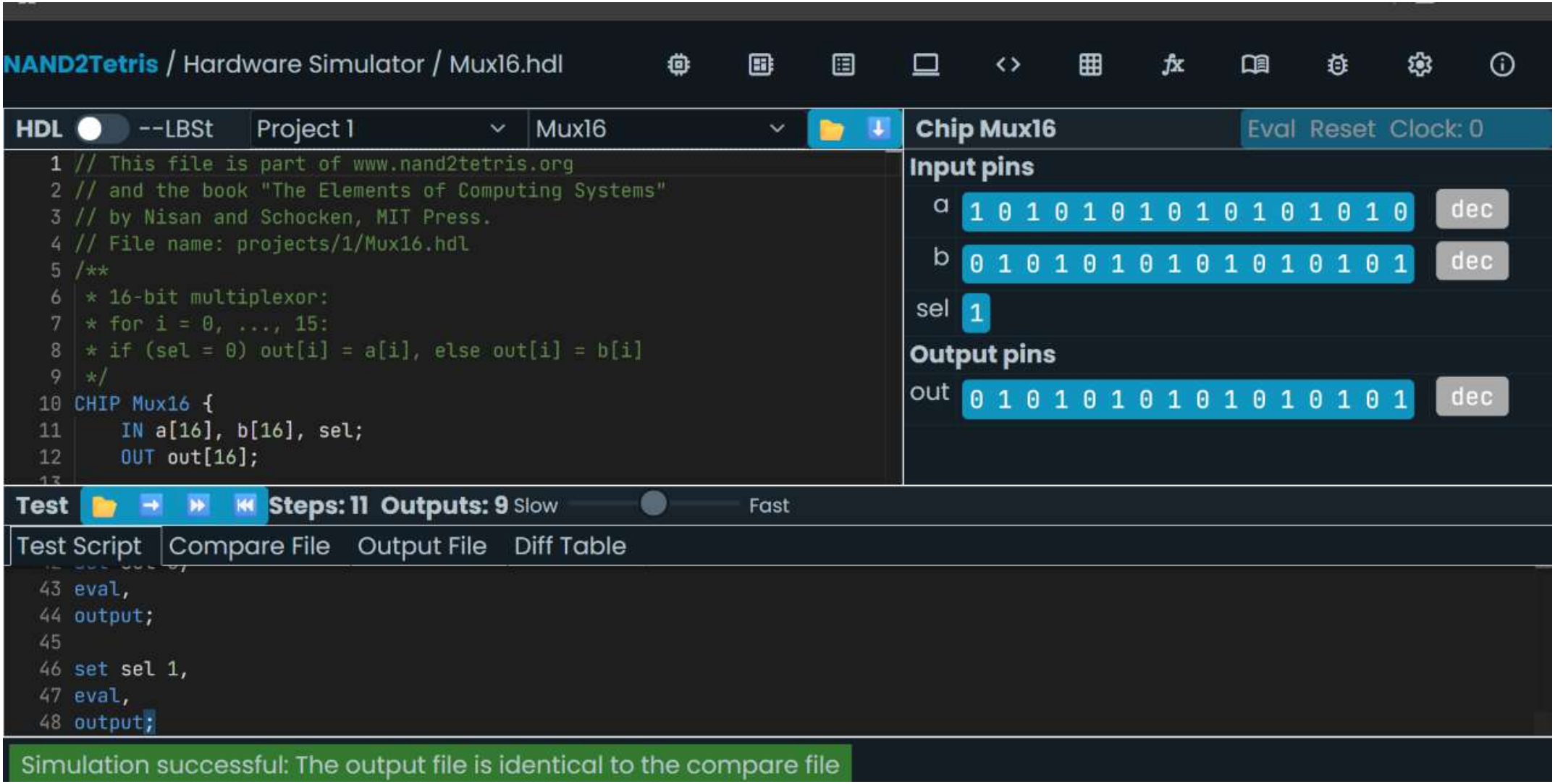

NAND2Tetris / Hardware Simulator / Mux.hdl
HDL --LBSt
Project 1
Mux
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Mux.hdl
5 /**
6 * Multiplexor:
7 * if (sel = 0) out = a, else out = b
8 */
9 CHIP Mux {
10 IN a, b, sel;
11 OUT out;
Chip Mux
Eval Reset Clock: 0
Input pins
a 1
b 1
sel 1
Output pins
out 1
Internal pins
xl 0
Test
Steps: 11 Outputs: 9 Slow
Fast
Test Script
Compare File
Output File
Diff Table
43 eval,
44 output;
45
46 set sel 1,
47 eval,
48 output;
Simulation successful: The output file is identical to the compare file

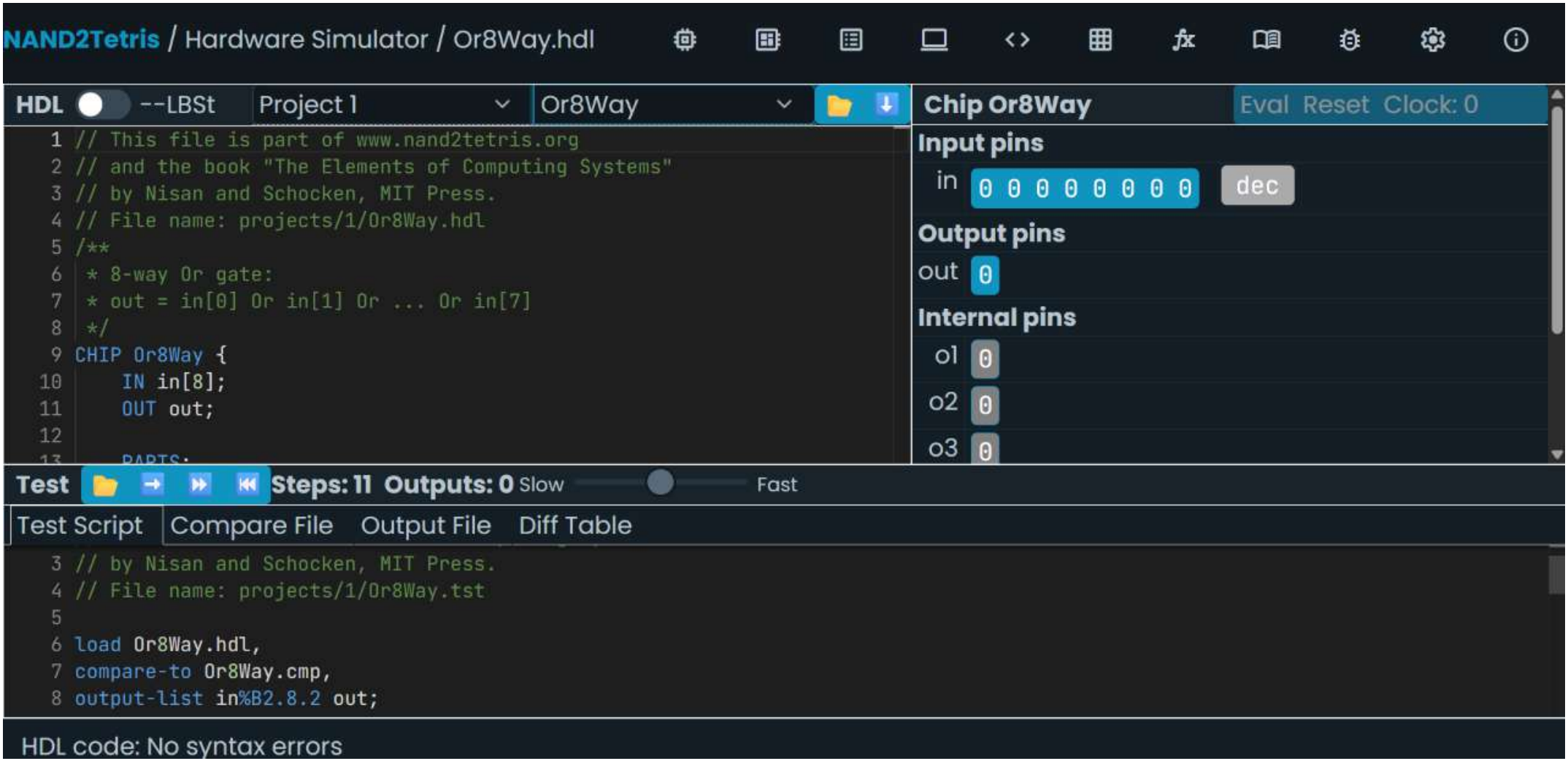

NAND2Tetris / Hardware Simulator / DMux.hdl
HDL --LBSt
Project 1
DMux
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/DMux.hdl
5 /**
6 * Demultiplexor:
7 * [a, b] = [in, 0] if sel = 0
8 * [0, in] if sel = 1
9 */
10 CHIP DMux {
11 IN in, sel;
12 OUT a, b;
Chip DMux
Eval Reset Clock: 0
Input pins
in 1
sel 1
Output pins
a 0
b 1
Internal pins
x1 0
Test
Steps: 7 Outputs: 5
Slow
Fast
Test Script
Compare File
Output File
Diff Table
21 eval,
22 output;
23
24 set sel 1,
25 eval,
26 output;
Simulation successful: The output file is identical to the compare file

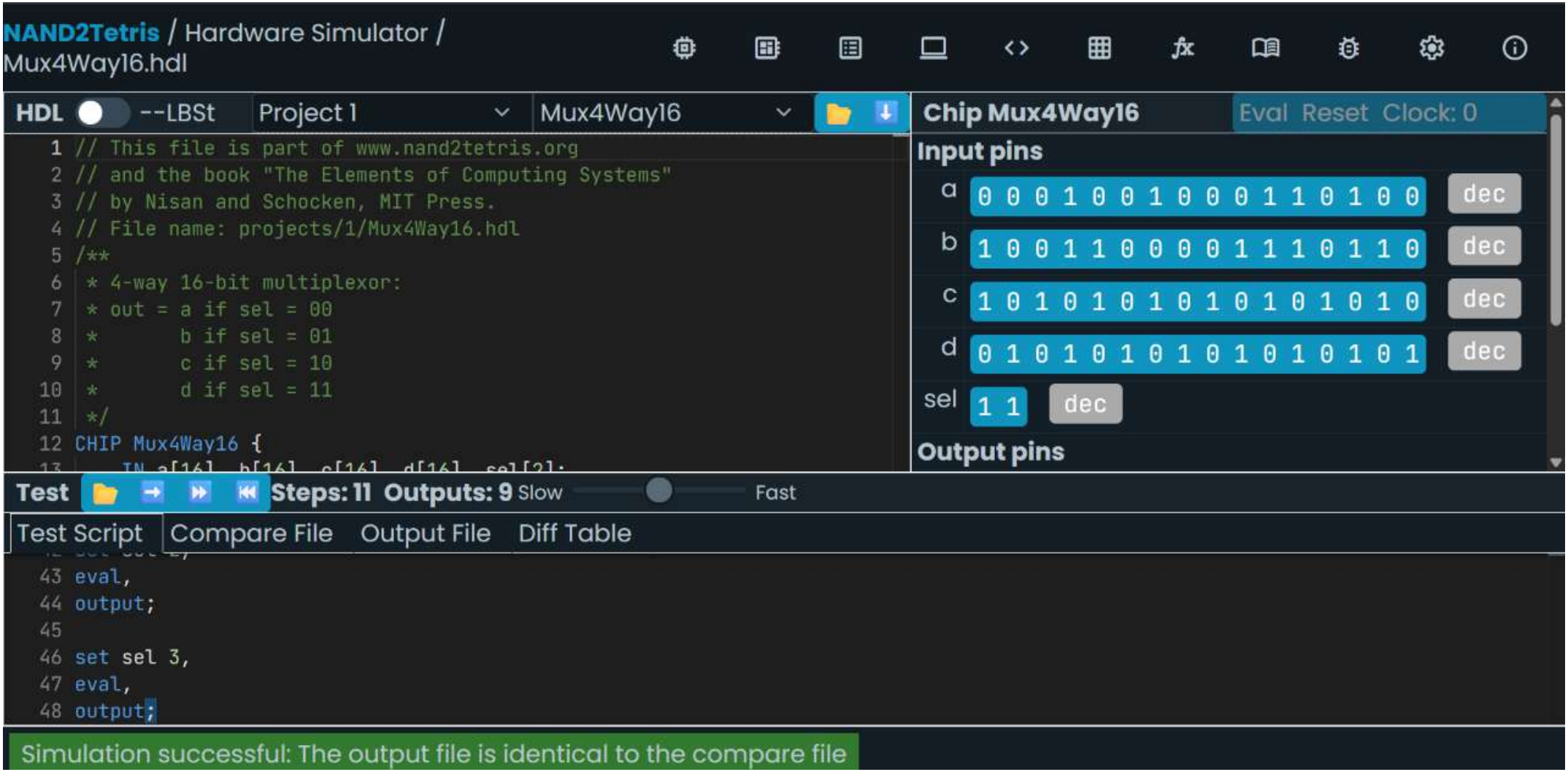

NAND2Tetris / Hardware Simulator / Not16.hdl
HDL --LBSt
Project 1
Not16
5 /**
8 * out[1] = Not(a[1])
9 */
10 CHIP Not16 {
11 IN in[16];
12 OUT out[16];
13
14 PARTS:
15 Not(in=in[0] , out=out[0]);
16 Not(in=in[1] , out=out[1]);
17 Not(in=in[2] , out=out[2]);
18 Not(in=in[3] , out=out[3]);
19 Not(in=in[4] , out=out[4]);
Chip Not16
Eval Reset Clock: 0
Input pins
in 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0
dec
Output pins
out 1 1 1 0 1 1 0 1 1 1 0 0 1 0 1 1
dec
Test
Steps: 8 Outputs: 6 Slow
Fast
Test Script
Compare File
Output File
Diff Table
23 eval,
24 output;
25
26 set in %B0001001000110100,
27 eval,
28 output;
Simulation successful: The output file is identical to the compare file

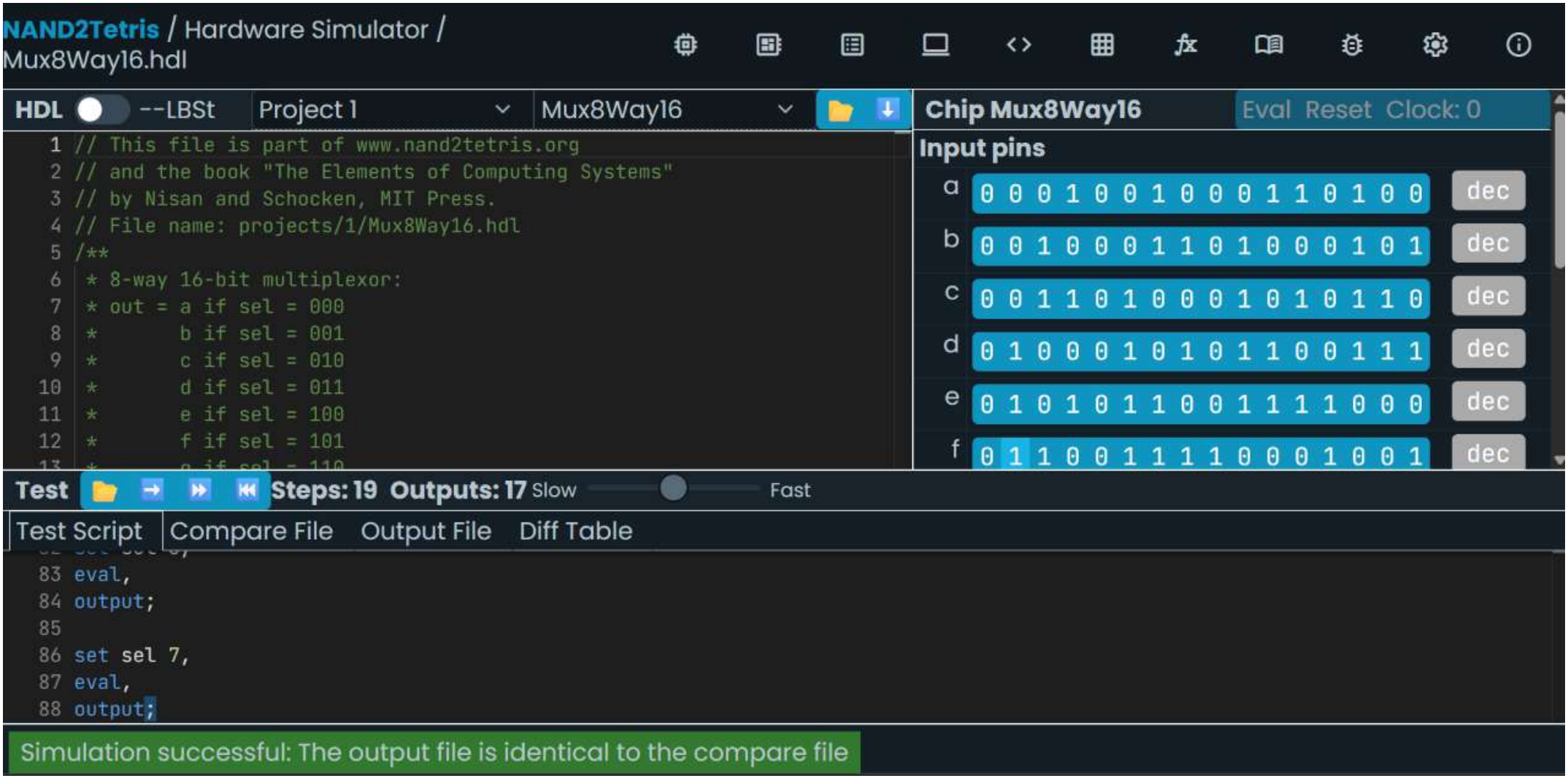

NAND2Tetris / Hardware Simulator / And16.hdl
HDL --LBSt
Project 1
And16
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/And16.hdl
5 /**
6 * 16-bit And gate:
7 * for i = 0, ..., 15:
8 * out[i] = a[i] And b[i]
9 */
10 CHIP And16 {
11 IN a[16], b[16];
12 OUT out[16];
Chip And16
Eval Reset Clock: 0
Input pins
a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0
dec
b 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0
dec
Output pins
out 0 0 0 1 0 0 0 0 0 0 1 1 0 1 0 0
dec
Test
Steps: 9 Outputs: 7
Slow
Fast
Test Script
Compare File
Output File
Diff Table
33 output;
34
35 set a %B0001001000110100,
36 set b %B1001100001110110,
37 eval,
38 output;
Simulation successful: The output file is identical to the compare file

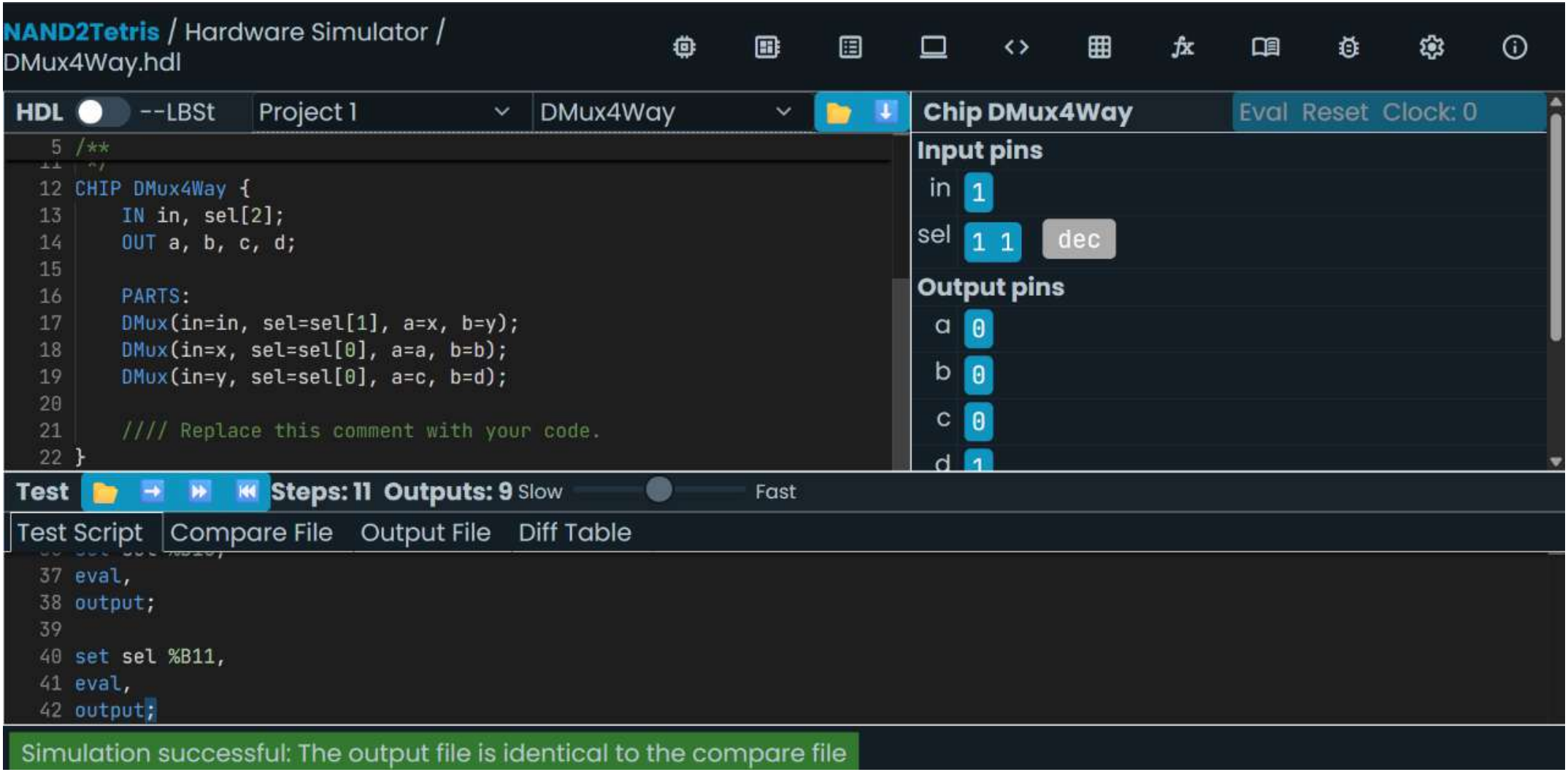

NAND2Tetris / Hardware Simulator / Or16.hdl
HDL --LBSt
Project 1
Or16
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Or16.hdl
5 /**
6 * 16-bit Or gate:
7 * for i = 0, ..., 15:
8 * out[i] = a[i] Or b[i]
9 */
10 CHIP Or16 {
11 IN a[16], b[16];
12 OUT out[16];
Chip Or16
Eval Reset Clock: 0
Input pins
a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0
dec
b 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0
dec
Output pins
out 1 0 0 1 1 0 1 0 0 1 1 1 0 1 1 0
dec
Test
Steps: 9 Outputs: 7 Slow
Fast
Test Script
Compare File
Output File
Diff Table
33 output;
34
35 set a %B0001001000110100,
36 set b %B1001100001110110,
37 eval,
38 output;
Simulation successful: The output file is identical to the compare file

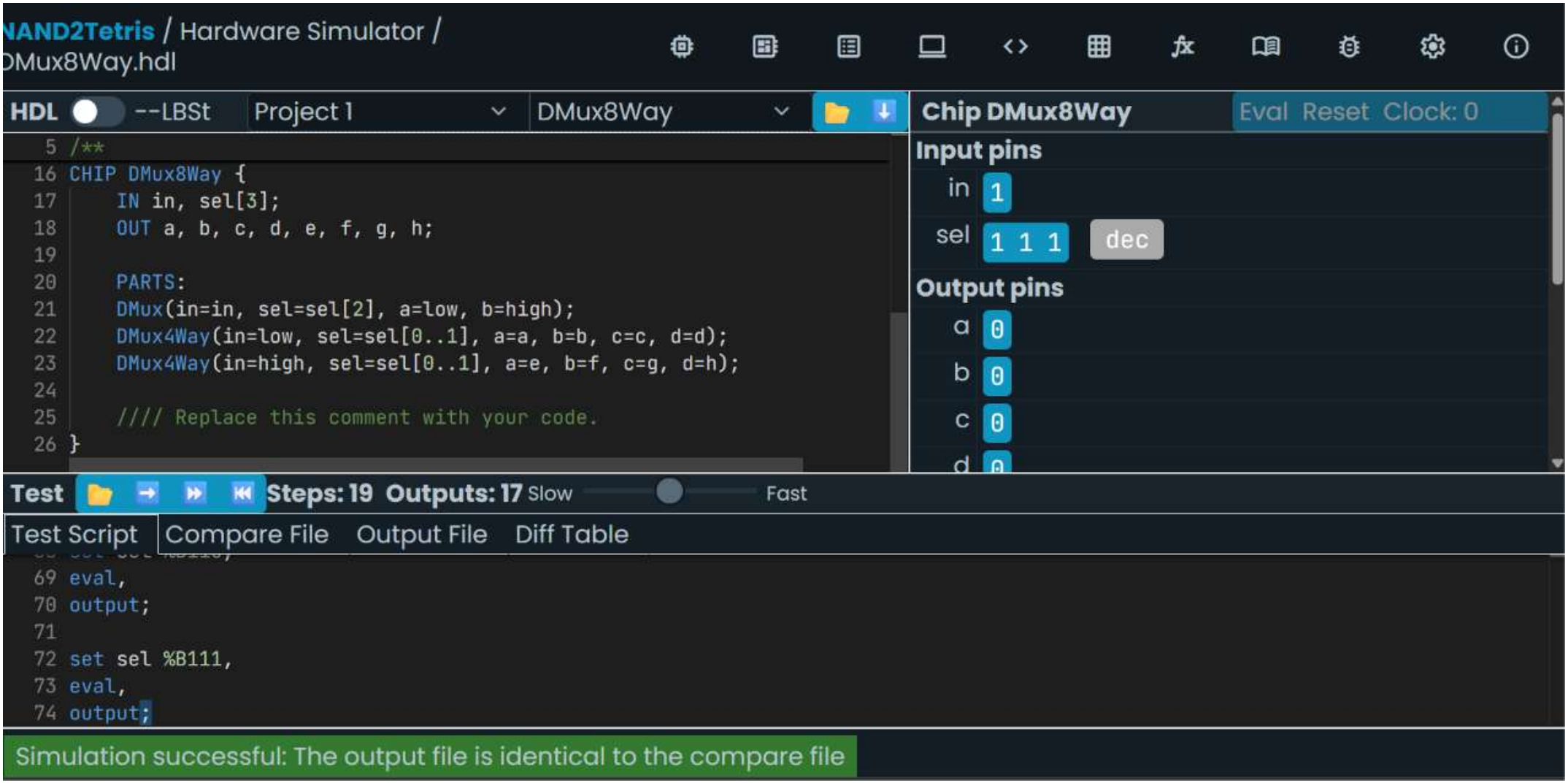

NAND2Tetris / Hardware Simulator / Mux16.hdl
HDL --LBSt
Project 1
Mux16
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Mux16.hdl
5 /**
6 * 16-bit multiplexor:
7 * for i = 0, ..., 15:
8 * if (sel = 0) out[i] = a[i], else out[i] = b[i]
9 */
10 CHIP Mux16 {
11 IN a[16], b[16], sel;
12 OUT out[16];
Chip Mux16
Eval Reset Clock: 0
Input pins
a 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0
dec
b 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
dec
sel 1
Output pins
out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
dec
Test
Steps: 11 Outputs: 9
Slow
Fast
Test Script
Compare File
Output File
Diff Table
43 eval,
44 output;
45
46 set sel 1,
47 eval,
48 output;
Simulation successful: The output file is identical to the compare file

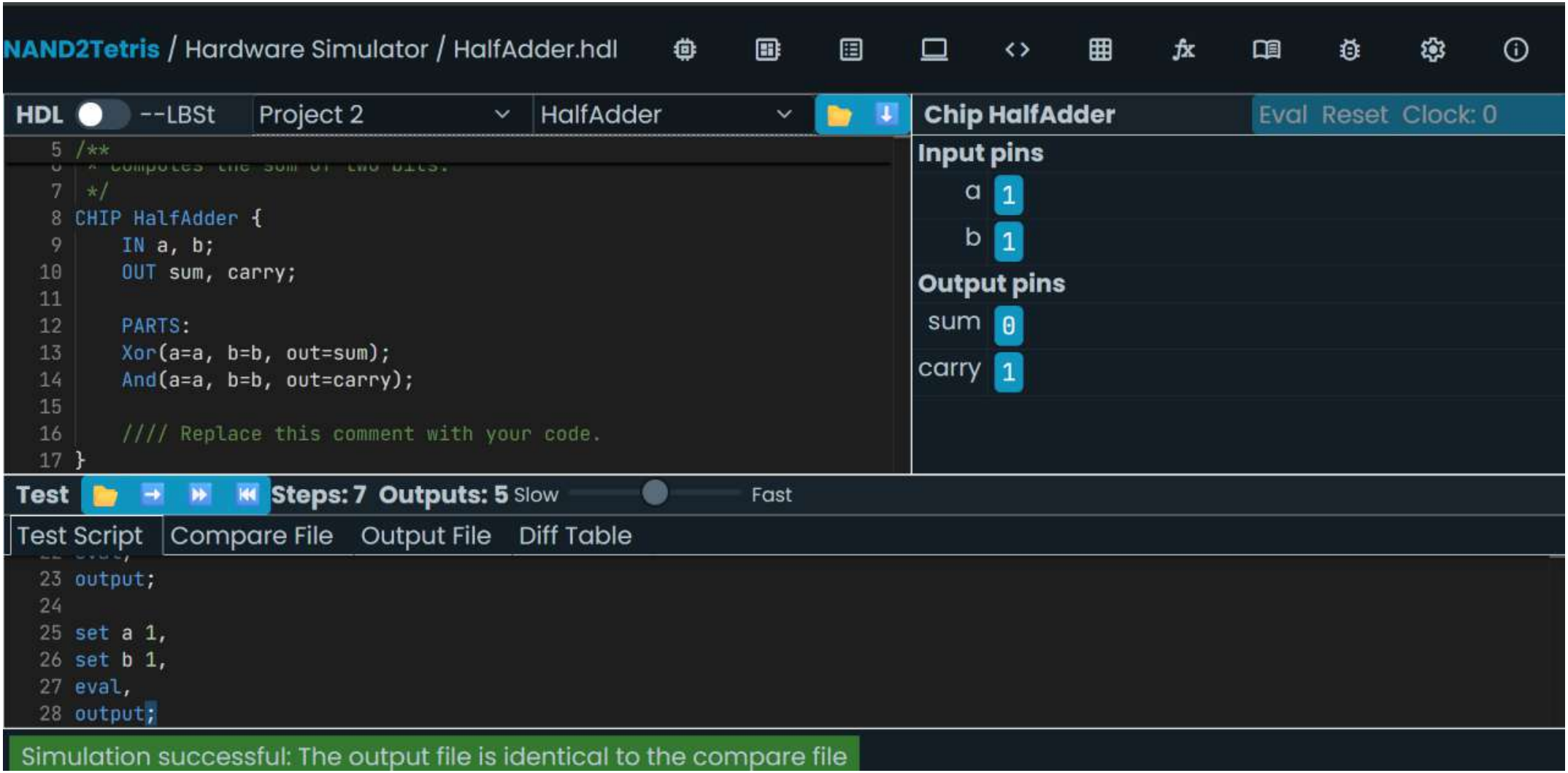

NAND2Tetris / Hardware Simulator / Or8Way.hdl
HDL --LBSt Project 1 Or8Way
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Or8Way.hdl
5 /**
6 * 8-way Or gate:
7 * out = in[0] Or in[1] Or ... Or in[7]
8 */
9 CHIP Or8Way {
10 IN in[8];
11 OUT out;
Chip Or8Way Eval Reset Clock: 0
Input pins
in 0 0 0 0 0 0 0 0 dec
Output pins
out 0
Internal pins
o1 0
o2 0
o3 0
Test Steps: 11 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Or8Way.tst
5
6 load Or8Way.hdl,
7 compare-to Or8Way.cmp,
8 output-list in%B2.8.2 out;
HDL code: No syntax errors

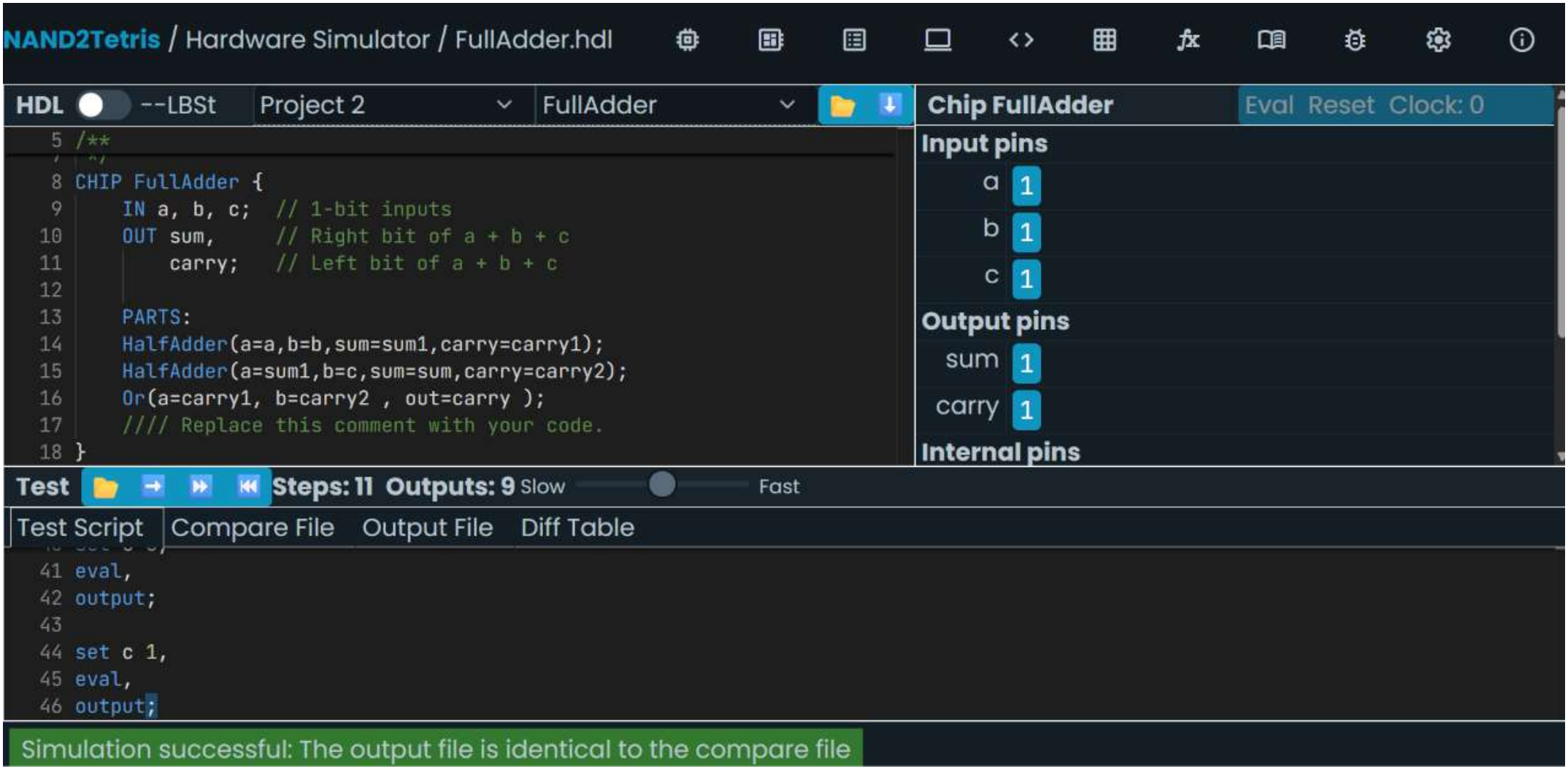

NAND2Tetris / Hardware Simulator /
Mux4Way16.hdl
HDL --LBSt
Project 1
Mux4Way16
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Mux4Way16.hdl
5 /**
6 * 4-way 16-bit multiplexor:
7 * out = a if sel = 00
8 * b if sel = 01
9 * c if sel = 10
10 * d if sel = 11
11 */
12 CHIP Mux4Way16 {
Chip Mux4Way16
Eval Reset Clock: 0
Input pins
a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec
b 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0 dec
c 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 dec
d 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
sel 1 1 dec
Output pins
Test
Steps: 11 Outputs: 9 Slow
Fast
Test Script
Compare File
Output File
Diff Table
43 eval,
44 output;
45
46 set sel 3,
47 eval,
48 output;
Simulation successful: The output file is identical to the compare file

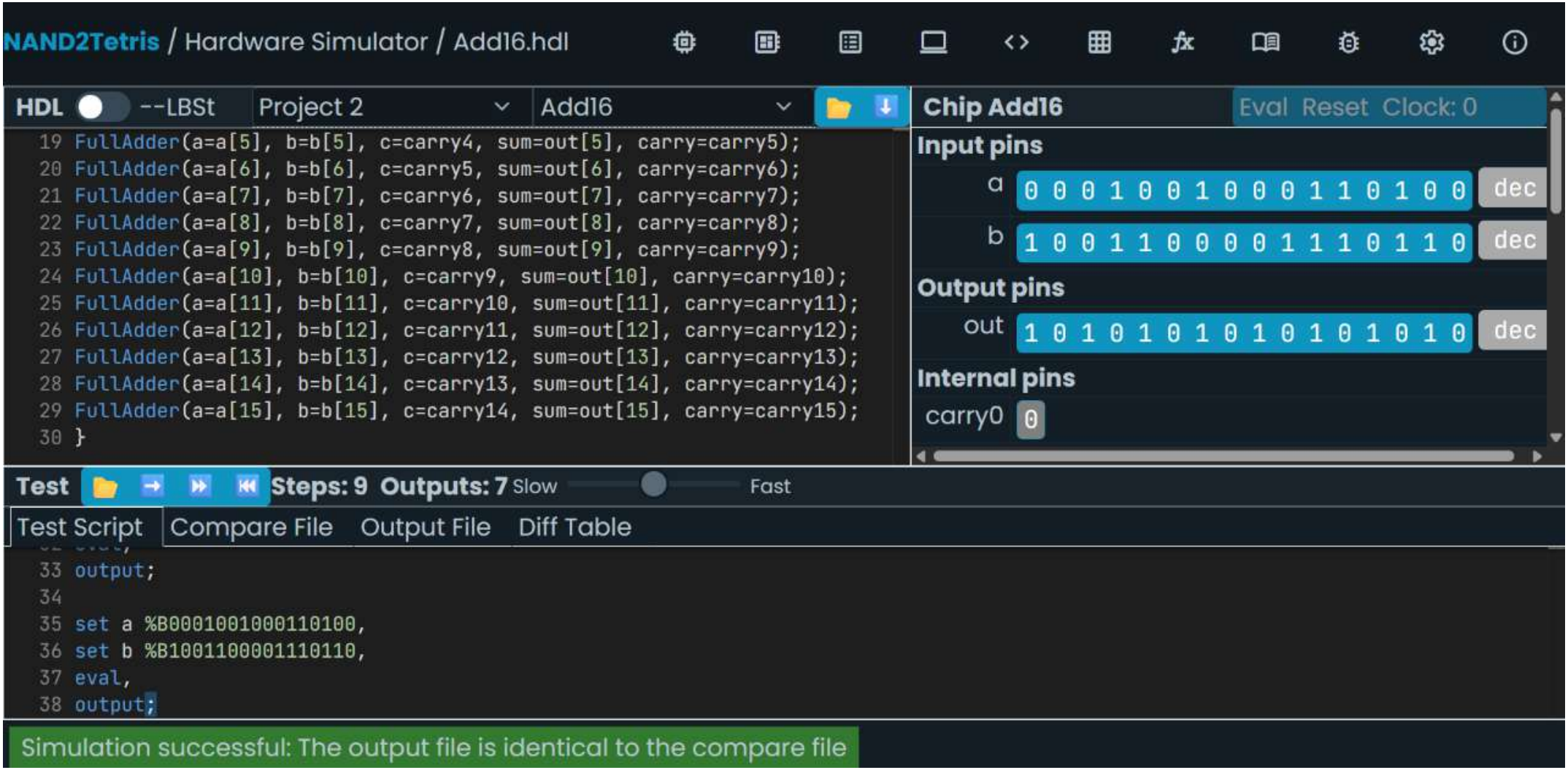

NAND2Tetris / Hardware Simulator /
Mux8Way16.hdl
HDL --LBSt
Project 1
Mux8Way16
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Mux8Way16.hdl
5 /**
6 * 8-way 16-bit multiplexor:
7 * out = a if sel = 000
8 * b if sel = 001
9 * c if sel = 010
10 * d if sel = 011
11 * e if sel = 100
12 * f if sel = 101
Chip Mux8Way16
Eval Reset Clock: 0
Input pins
a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec
b 0 0 1 0 0 0 1 1 0 1 0 0 0 1 0 1 dec
c 0 0 1 1 0 1 0 0 0 1 0 1 0 1 1 0 dec
d 0 1 0 0 0 1 0 1 0 1 1 0 0 1 1 1 dec
e 0 1 0 1 0 1 1 0 0 1 1 1 1 0 0 0 dec
f 0 1 1 0 0 1 1 1 1 0 0 0 1 0 0 1 dec
Test
Steps: 19 Outputs: 17 Slow
Fast
Test Script
Compare File
Output File
Diff Table
83 eval,
84 output;
85
86 set sel 7,
87 eval,
88 output;
Simulation successful: The output file is identical to the compare file

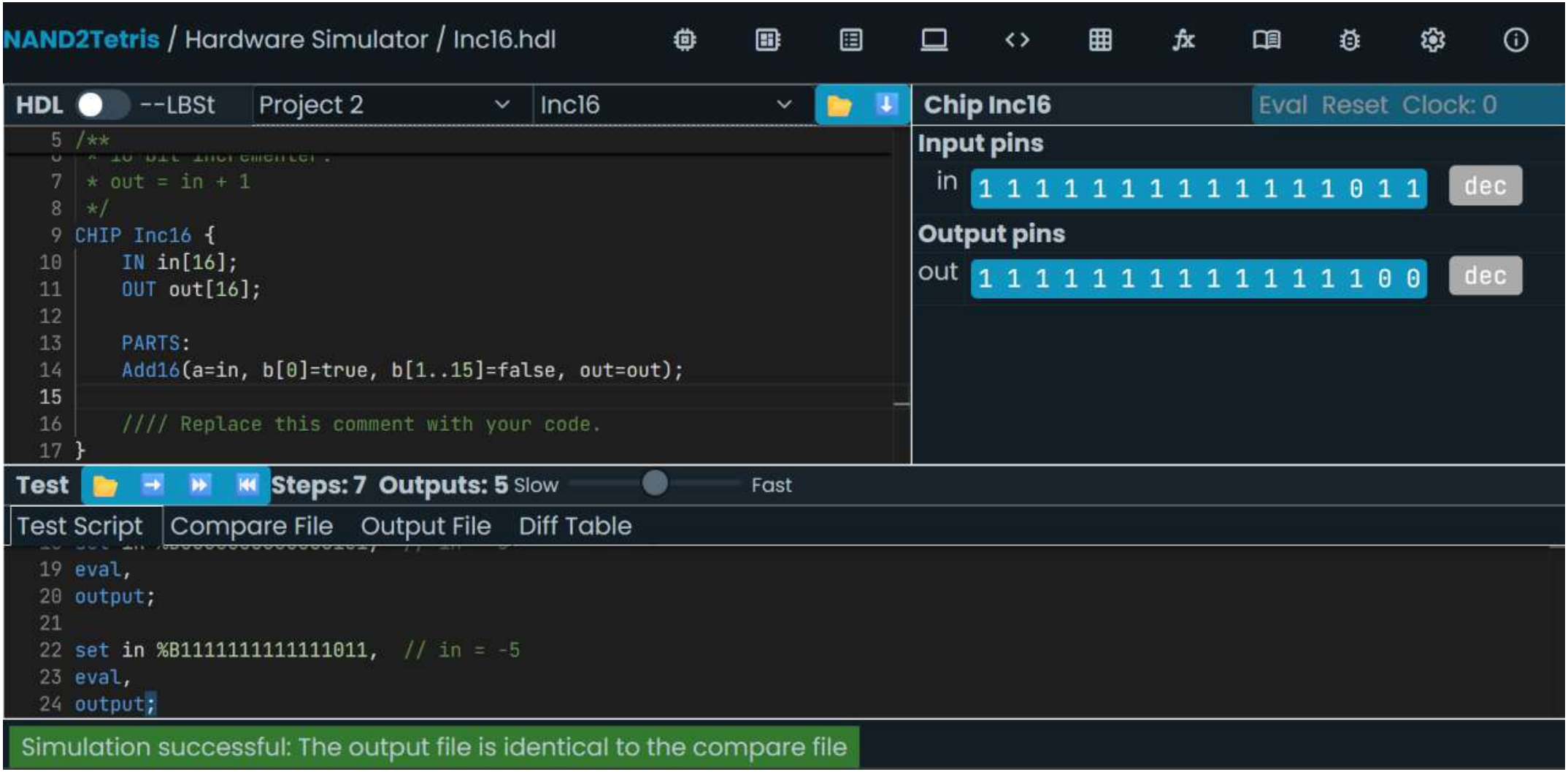

NAND2Tetris / Hardware Simulator /
DMux4Way.hdl
HDL --LBSt
Project 1
DMux4Way
5 /**
12 CHIP DMux4Way {
13 IN in, sel[2];
14 OUT a, b, c, d;
16 PARTS:
17 DMux(in=in, sel=sel[1], a=x, b=y);
18 DMux(in=x, sel=sel[0], a=a, b=b);
19 DMux(in=y, sel=sel[0], a=c, b=d);
21 //// Replace this comment with your code.
22 }
Chip DMux4Way
Eval Reset Clock: 0
Input pins
in 1
sel 1 1 dec
Output pins
a 0
b 0
c 0
Test
Steps: 11 Outputs: 9 Slow Fast
Test Script
Compare File
Output File
Diff Table
37 eval,
38 output;
40 set sel %B11,
41 eval,
42 output;
Simulation successful: The output file is identical to the compare file

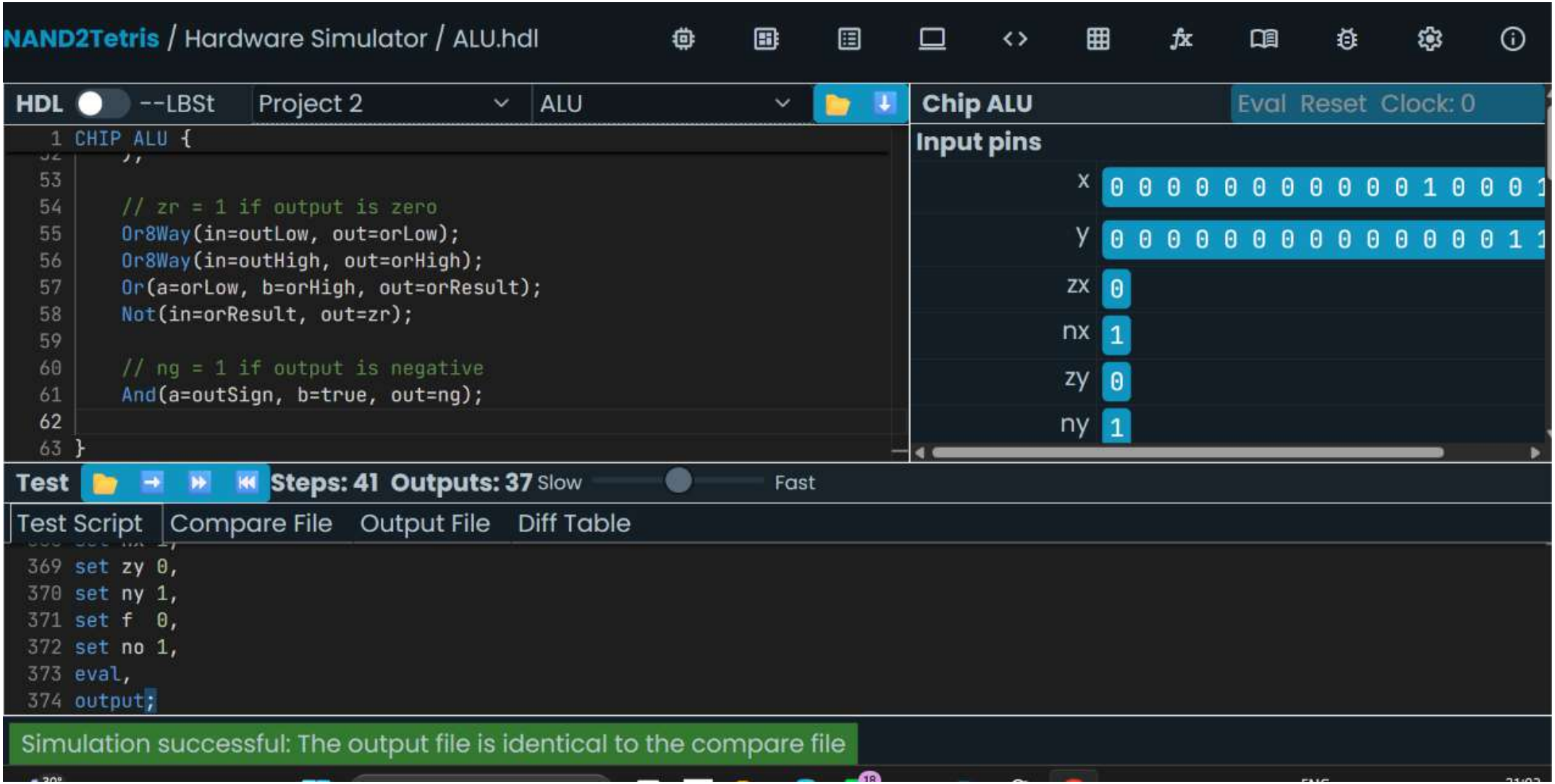

NAND2Tetris / Hardware Simulator /
DMux8Way.hdl
HDL --LBSt
Project 1
DMux8Way
5 /**
16 CHIP DMux8Way {
17 IN in, sel[3];
18 OUT a, b, c, d, e, f, g, h;
19
20 PARTS:
21 DMux(in=in, sel=sel[2], a=low, b=high);
22 DMux4Way(in=low, sel=sel[0..1], a=a, b=b, c=c, d=d);
23 DMux4Way(in=high, sel=sel[0..1], a=e, b=f, c=g, d=h);
24
25 //// Replace this comment with your code.
26 }
Chip DMux8Way
Eval Reset Clock: 0
Input pins
in 1
sel 1 1 1
dec
Output pins
a 0
b 0
c 0
Test
Steps: 19 Outputs: 17 Slow
Fast
Test Script
Compare File
Output File
Diff Table
69 eval,
70 output;
71
72 set sel %B111,
73 eval,
74 output;
Simulation successful: The output file is identical to the compare file


NAND2Tetris / Hardware Simulator / HalfAdder.hdl
HDL --LBSt Project 2 HalfAdder
5 /**
7 */
8 CHIP HalfAdder {
9 IN a, b;
10 OUT sum, carry;
12 PARTS:
13 Xor(a=a, b=b, out=sum);
14 And(a=a, b=b, out=carry);
16 //// Replace this comment with your code.
17 }
Chip HalfAdder
Eval Reset Clock: 0
Input pins
a 1
b 1
Output pins
sum 0
carry 1
Test
Steps: 7 Outputs: 5
Slow
Fast
Test Script
Compare File
Output File
Diff Table
23 output;
25 set a 1,
26 set b 1,
27 eval,
28 output;
Simulation successful: The output file is identical to the compare file


NAND2Tetris / Hardware Simulator / FullAdder.hdl
HDL --LBSt Project 2 FullAdder
5 /**
8 CHIP FullAdder {
9 IN a, b, c; // 1-bit inputs
10 OUT sum, // Right bit of a + b + c
11 carry; // Left bit of a + b + c
13 PARTS:
14 HalfAdder(a=a,b=b,sum=sum1,carry=carry1);
15 HalfAdder(a=sum1,b=c,sum=sum,carry=carry2);
16 Or(a=carry1, b=carry2 , out=carry);
17 //// Replace this comment with your code.
18 }
Chip FullAdder Eval Reset Clock: 0
Input pins
a 1
b 1
c 1
Output pins
sum 1
carry 1
Internal pins
Test Steps: 11 Outputs: 9 Slow Fast
Test Script Compare File Output File Diff Table
41 eval,
42 output;
44 set c 1,
45 eval,
46 output;
Simulation successful: The output file is identical to the compare file


NAND2Tetris / Hardware Simulator / Add16.hdl
HDL --LBSt Project 2 Add16
19 FullAdder(a=a[5], b=b[5], c=carry4, sum=out[5], carry=carry5);
20 FullAdder(a=a[6], b=b[6], c=carry5, sum=out[6], carry=carry6);
21 FullAdder(a=a[7], b=b[7], c=carry6, sum=out[7], carry=carry7);
22 FullAdder(a=a[8], b=b[8], c=carry7, sum=out[8], carry=carry8);
23 FullAdder(a=a[9], b=b[9], c=carry8, sum=out[9], carry=carry9);
24 FullAdder(a=a[10], b=b[10], c=carry9, sum=out[10], carry=carry10);
25 FullAdder(a=a[11], b=b[11], c=carry10, sum=out[11], carry=carry11);
26 FullAdder(a=a[12], b=b[12], c=carry11, sum=out[12], carry=carry12);
27 FullAdder(a=a[13], b=b[13], c=carry12, sum=out[13], carry=carry13);
28 FullAdder(a=a[14], b=b[14], c=carry13, sum=out[14], carry=carry14);
29 FullAdder(a=a[15], b=b[15], c=carry14, sum=out[15], carry=carry15);
30 }
Chip Add16
Eval Reset Clock: 0
Input pins
a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec
b 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0 dec
Output pins
out 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 dec
Internal pins
carry0 0
Test
Steps: 9 Outputs: 7
Slow
Fast
Test Script
Compare File
Output File
Diff Table
33 output;
34
35 set a %B0001001000110100,
36 set b %B1001100001110110,
37 eval,
38 output;
Simulation successful: The output file is identical to the compare file


NAND2Tetris / Hardware Simulator / Inc16.hdl
HDL --LBSt
Project 2
Inc16
5 /**
7 * out = in + 1
8 */
9 CHIP Inc16 {
10 IN in[16];
11 OUT out[16];
13 PARTS:
14 Add16(a=in, b[0]=true, b[1..15]=false, out=out);
16 //// Replace this comment with your code.
17 }
Chip Inc16
Eval Reset Clock: 0
Input pins
in 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1
dec
Output pins
out 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0
dec
Test
Steps: 7 Outputs: 5
Slow
Fast
Test Script
Compare File
Output File
Diff Table
19 eval,
20 output;
22 set in %B1111111111111011, // in = -5
23 eval,
24 output;
Simulation successful: The output file is identical to the compare file


NAND2Tetris / Hardware Simulator / ALU.hdl
HDL --LBSt Project 2 ALU
1 CHIP ALU {
53
54 // zr = 1 if output is zero
55 Or8Way(in=outLow, out=orLow);
56 Or8Way(in=outHigh, out=orHigh);
57 Or(a=orLow, b=orHigh, out=orResult);
58 Not(in=orResult, out=zr);
59
60 // ng = 1 if output is negative
61 And(a=outSign, b=true, out=ng);
62
63 }
Chip ALU
Eval Reset Clock: 0
Input pins
x 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 1
y 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1
zx 0
nx 1
zy 0
ny 1
Test
Steps: 41 Outputs: 37 Slow Fast
Test Script
Compare File
Output File
Diff Table
369 set zy 0,
370 set ny 1,
371 set f 0,
372 set no 1,
373 eval,
374 output;
Simulation successful: The output file is identical to the compare file